# Joey Chung

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## EDUCATION

**University of Michigan, Ann Arbor** *Aug. 2025 – Present*
B.S. in Chemistry | GPA: 3.99 / 4.00

**University of Wisconsin, Madison** *Sept. 2022 – May 2024*
Coursework toward B.S. in Food Science | GPA: 3.96 / 4.00
Honors: Dean's List (4 semesters); Norman F. Olson Family Scholarship (2024); Bill Reese Dairy, Deli & Bakery Study Grant (2023); Wisconsin Alumni Association Korean Chapter Scholarship (2022)

## RESEARCH & TEACHING

**Maldonado Group | Undergraduate Research Assistant** *UM-Ann Arbor | April 2026 – Present*
*Advisor: Prof. Stephen Maldonado | Responsible for the group's spectroscopic ellipsometer and AFM*

- Measure thin film thickness by spectroscopic ellipsometry (HORIBA UVISEL II), building layer models in DeltaPsi2 and judging each fit by $\chi^2$; calibrate on an aluminum mirror standard and verify against a 100 nm thermal oxide standard.
- Characterize surface morphology and step heights by tapping mode AFM (Veeco MultiMode, NanoScope Analysis) on own and group members' samples; recalibrated the Z axis after a 100 nm step standard read 10% high.
- Fabricate $Al_2O_3$ films on n-type Si(111) from bare wafer: solvent degreasing, piranha cleaning, $NH_4F$/HF oxide etching, plasma treatment, and thermal ALD (TMA/$H_2O$, Anric AT410) with cycle count set to target thickness.

**Organic Chemistry Peer Mentor | Academic Support** *UW-Madison | Fall 2023 – Spring 2024*

- Resolved 100+ academic inquiries on Piazza and led interactive sessions on organic chemistry concepts and multistep reaction mechanisms.

## PROFESSIONAL EXPERIENCE

**DAEYANG Co. Ltd | Research Assistant** *Daegu, South Korea | Sept. 2024 – May 2025*
*Corporate research institute developing thermoelectric materials and modules for energy applications (Website)*

- Ran medium phosphorus (6 to 8% P) electroless nickel plating on bismuth telluride pellets, with alcohol-based cleaning and pH control. After spotting a plating stability issue, proposed and designed a custom titanium holding jig with drawings and specifications; the build stalled on scheduling, fabricator sourcing, and MOQ constraints.
- Inspected diced legs against design tolerances and classified defects (machining error, burring, asymmetry). On a Korea Electrotechnology Research Institute (KERI) module build, separately ran incoming inspection and tray loading of supplied legs, pulling nonconforming units before SMT.
- Supported senior engineers on SMT assembly of 8×8 modules (128 legs each): solder paste prep, frame setup, substrate loading, and lead wire soldering after reflow. Output ran 20 to 30 modules a week, passing 50 during government R&D cycles.
- Ran final electrical inspection (resistance and voltage against spec), logging results in Excel for traceability and flagging units for retest.
- Collected generator performance data across five cooling configurations (forced and natural air, forced and natural water, antifreeze).

**Military Service | Auxiliary Police Officer (Sergeant)** *Pohang, South Korea | Nov. 2019 – May 2021*
*Mandatory service under the Ministry of National Defense and the Korean National Police Agency*

- Led a squad of 12 officers for 7 of the 18-month term through protest response and missing person searches across mountainous and coastal terrain; located 3 missing individuals and escorted them to safety, earning an official commendation.
- Managed police lines during civil disturbances and took part in joint readiness drills with the ROK Marine Corps, including chemical threat response protocols.

## LEADERSHIP & GLOBAL ENGAGEMENT

**Michigan Rifle Team | Member** *UM-Ann Arbor | Sept. 2025 – Present*

- Compete in three-position and air rifle events (Camp Perry Open, WIRC Championship); create team photo content and organize events.

**Food Science Club | Member** *UW-Madison | Sept. 2022 – Dec. 2023*

- Coordinated HR-led recruitment sessions with industry representatives and organized cooking activities for members.

**International Food and Cultural Engagement | Cultural Outreach** *8+ Countries | Jan. 2021 – Present*

- Cooked Korean fusion dishes (e.g., bulgogi tacos, gochujang sauces) for friends and communities at informal and festival style events in Spain, Poland, Mexico, and more.

## SKILLS

- **Process & Quality:** SMT workflow, solder paste prep, fixture design, dimensional inspection, defect classification, data logging and traceability.
- **Thin Film & Surface Characterization:** spectroscopic ellipsometry (DeltaPsi2 modeling), tapping mode AFM (NanoScope Analysis), thermal ALD, instrument calibration.
- **Wet Chemistry & Wafer Prep:** piranha and $NH_4F$/HF etching, solvent degreasing, plasma treatment, electroless plating, chemical safety.